POST

# Fundamental?

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Quantum physics?

---“Is quantum physics fundamental?

… I highly doubt it.”

- Jed Anderson, EnviroAI

“Aristotelian and Newtonian physics wasn’t found to be fundamental …

… just 1universe in the Universe?

… “I highly doubt it.”

- Jed Anderson

… does anything underly quantum physics?

… “I highly think so.”

- Jed Anderson

FUNDAMENTAL PHYSICS?

The continual iconoclastic evolution in our cosmological understanding …

- Aristotelian Physics (345 BC)
- Universe is made of 5 elements (https://r20.rs6.net/tn.jsp?f=001BQyiS-xnM33eozdnkt_dd6nSRSkCfn9h5lril9657rpp7-nmhJAsA7aJ_CSUg1YojaFBWb0msyc1kwqZhkomSGjz2a964Pa9DIMljtmdc2WuGG6ftkFZPfzViL700GBX00PJGH-ipAAEKu__QYi08BD85nEDk28kRF6TCkoN-6TOpfNEkTKO7Q==&c=&ch=): earth, air, fire, water, and aether. Objects have natural motions: those of earth and water tend to fall; those of air and fire, to rise.
- Galilean Physics (1592)
- Galileo determines first laws of motion by experimentation
- Develops the concepts of: velocity, force, and inertia
- natural state of motion is uniform motion or rest
- objects fall at same rate regardless of mass
- Keplerian Physics (1609)
- Kepler produces first kinematic description of orbits
- Each object moves in an elliptical orbit
- Newtonian Physics ("Classical Physics" (1687)
- Newton provides dynamical description of universe, using:

1. acceleration
2. momentum

3. conservation laws

[Jed description: Everything on an xyz axis in space-time]

- Quantum Physics (“Modern Physics) (1931)
- Particles are waves and vice-versa
- Probability is all we ever know
- Measurement determines reality

[Jed description: Everything is probabilities]

- _____? (“Post-Modern” Physics)(_____)

Reverse Engineering the Universe

---“I think physics is about done with reductionism. It’s taken us a long way, but the next jump in physics I think will unleash much higher levels in imaginative heuristics.”- Jed Anderson

---“As usual, nature’s imagination far surpasses our own, as we have seen from the other theories which are subtle and deep.” - Richard Feynman

FUNDAMENTAL UNIVERSE?

The continual iconoclastic evolution in our cosmological understanding …

- Our Earth is round not flat (this notion of a round earth is contrary to intuition and most of our everyday experience) - 500 B.C. (Pythagorus) [Eratosthenes calculated circumference in 240 B.C. and of course Columbus didn’t sail off the

edge of the earth when he tried it almost 2000 years later]

- Our Earth isn't the center of the universe[earth rotates around the sun] - 240 B.C. (Aristarchus) [Calculations by Copernicus in 1532 and astronomical observations by Galileo in 1632]
- Our sun isn't the center of the universe[stars are other suns] - 450 B.C. (Anaxagoras) [Bruno in 1584 and then Angelo Secchi proved through spectroscopy in 1860]
- Our galaxy isn't the center of the universe - 964 (Azophi) [Later Kant, Messier, Shapley, and finally Edwin Hubble … January 1, 1925"The day we discovered the universe"]
- Our universe isn't the center of the Universe [or Multiverse]- _____?

… "To think that our big bang was the only bang would seem at this point to me to be a bit narrow-minded, anthropocentric, andin contradiction to the continual iconoclastic evolution in our cosmological understanding."

- Jed Anderson, EnviroAI

How many universes?Interview (https://r20.rs6.net/tn.jsp?f=001BQyiS-xnM33eozdnkt_dd6nSRSkCfn9h5lril9657rpp7-nmhJAsAwbLYP9xKdMMDG63KPfPUIjNTJuuHsYDIHsk632BfgqEiINpeLNxcGjKG9v4RdH0rWbK4tLoyogUMkyomPoS0N-XbEztf0I3FQ==&c=&ch=) with Steven Weinberg, recently deceased University of Texas Professor and Nobel Prize Winner in Physics.

How many universes? Interview (https://r20.rs6.net/tn.jsp?f=001BQyiS-xnM33eozdnkt_dd6nSRSkCfn9h5lril9657rpp7-nmhJAsAyqsepah_UCarkRP-5OxC3-HTuC63300K7cwSUIrupo3jFdHqcK3kMSLbhMdJhShqWdYbN4rV5c6ZuM2UMpT7-dtZNLfA8sUn5GIrLDGQAPHBXuHCn7I2l99b-dWp-DOafprNmrOxyiA3sDMk9XllFnGFZkLgmVV7A==&c=&ch=) with Max Tegmark, MIT Professor of Physics.

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